



AI-assisted automated scoring of picture-cued writing tasks for language assessment

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Abstract

Grading assignments is inherently subjective and time-consuming; automatic scoring tools can greatly reduce teacher workload and shorten the time needed for providing feedback to learners. The purpose of this paper is to propose a novel method for automatically scoring student responses to picture-cued writing tasks. As a popular paradigm for language instruction and assessment, a picture-cued writing task typically requires students to describe a picture or pictures. Correspondingly, the automatic scoring methods must measure the link(s) between visual pictures and their textual descriptions. For this purpose, we first designed a picture-cued writing test and collected nearly 4 k responses from 279 K12 students. Based on these responses, we then developed an AI scoring model by incorporating the emerging cross-modal matching technology and some NLP algorithms. The performance of the model was evaluated carefully with six popular measures and was found to demonstrate accurate scoring results with a small mean absolute error of 0.479 and a high adjacent-agreement rate of 90.64%. We believe this method could reduce the subjective elements inherent in human grading and save teachers' time from the mundane task of grading to other valuable endeavors such as designing teaching plans based on AI-generated diagnosis of student progress.

Keywords Automated writing assessment · Picture-cued writing · Cross-modal matching · Artificial intelligence

1 Introduction

In response to an ongoing need for innovative assessment in language writing instruction, researchers and educators are increasingly turning to computer-based approaches, especially automated writing assessment (Palermo and Thomson, 2018; Woodworth & Barkaoui, 2020). By scoring the written text and analyzing its

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structure and grammar automatically, technological tools have the capability to provide immediate qualitative and quantitative feedback to students and teachers (Khoii & Doroudian, 2014), thereby, reducing teacher workload and shortening the time needed for providing feedback to students. At present, these tools are reshaping the way that people teach and learn language, especially through online courses and mobile apps that enable personalized autonomous learning.

Meanwhile, the popularization of multimodal educational data is also affecting modern language instruction in a profound way. Multimodal data could represent learning materials in multiple modalities, such as image, text, audio, and video, providing educators with more semiotic resources than traditional text-centered materials. Multimodal data also enable researchers to explore innovative learning practices that could fully utilize human cognitive capacities (Mangaroska et al., 2021) and pave the way for multimodal pedagogy (Erfanian Mohammadi et al., 2019). Picture-cued writing tasks widely used to assess K12 students is a good example of the application of multimodal data in education. The task asks students to describe a given picture or pictures. Therefore, both visual pictures and textual responses need to be considered simultaneously in assessing the responses. However, most of the existing research on automated writing assessment focuses on traditionally text-based writing and remains inadequate for picture-cued writing tasks. This is because the NLP-driven methods are unable to match textual responses with visual prompts. Fortunately, the recent emergence of cross-modal deep learning technology in the field of artificial intelligence provides an opportunity to fill in this gap. The technology allows researchers to extract and fuse information from multimodal data synchronously.

Utilizing this novel technology, this study developed and validated a method for automatically scoring student responses to picture-cued writing tasks; the method combined techniques from both cross-modal deep learning and natural language processing. The effectiveness and robustness of this innovative method were validated in various conditions, such as pictures with different characteristics and students in various subgroups. Our research was guided by the following research questions:

RQ1. How effectively can the cross-modal deep learning technology be used to measure the matching degrees between visual pictures and their responses written by k12 students?

RQ2. Can an AI-assisted tool be further developed for scoring student responses to picture-cued writing tasks automatically by incorporating cross-modal AI technology and NLP algorithms?

2 Background

2.1 Related research on automated writing evaluation

Numerous researches in education have established the importance of timely instructional feedback and its impact on learning (Hattie & Timperley, 2007; Kingston & Nash, 2011), especially on writing outcomes. Appropriate and detailed feedback can help students to understand whether their understanding

is on the right track and ways to improve (Shim, 2013). However, sufficient and high-quality feedback implies an enormous workload for teachers, especially in contexts where large classes are the norm. Grading and writing feedback to a large number of student writings is not only labor-intensive and time-consuming but also, at times, problematic and difficult to maintain consistency (Chapelle et al., 2015; Warschauer & Grimes, 2008; Wilson & Czik, 2016). Thanks to recent technological advancements, tools for automated writing evaluation (AWE) have been developed to grade and generate instantaneous diagnostic feedback, and consistent explanations for student writings (Khoii & Doroudian, 2014). These tools not only reduce teachers' workload and redirect their time to spend on more complex pedagogical tasks such as designing lessons to cater for learner diversity and to motivate learning (Link et al., 2020; Wilson & Czik, 2016). In terms of writing, AWE tools can assist the writing process and support their autonomous learning activities by providing personalized feedback along the way (Deeva et al., 2021), which is infeasible in traditional language classrooms. Existing research found these tools help students improve their writing quality (Palermo & Thomson, 2018; Stevenson & Phakiti, 2014), their ability to effectively revise their writing (Wang et al., 2020), their writing motivation (Wilson & Czik, 2016), their attitudes towards writing (Roscoe et al., 2018), and their writing self-efficacy (Wilson & Roscoe, 2020).

As AI-driven tools, the functions and performances of AWE are heavily dependent on the advances and selection of artificial intelligence technologies. The tools originally emerged as a means of assessing student writing in large-scale standardized testing contexts, and the most familiar tools include Educational Testing Service's e-Rater (Attali & Burstein, 2006), Vantage Learning's Intelli-Metric (Elliott et al., 2003), and Project Essay Grade (PEG) (now MI Write; Page, 2003). These tools usually assign scores in a two-step procedure automatically. In Step 1, natural language processing algorithms are used to extract semantic, syntactic, or rhetorical/discourse-level features related to writing quality. In Step 2, statistical or machine-learning algorithms are utilized to generate summative numeric scores as well as formative feedback based on patterns observed among the features (McCarthy et al., 2022; Strobl et al., 2019). Project Essay Grade (PEG), for instance, used a simple linear regression model based on features such as essay length and proposition counts to perform text classification tasks. Scoring tools like PEG heavily depend on feature engineering and knowledge of linguistic discourse (Lu & Cutumisu, 2021).

Inspired by recent advances in deep learning algorithms and word embedding techniques, a substantial body of research has emerged and contributed to improving the performance of automated scoring tools. In contrast to hand-crafted feature-based approaches, the systems are implemented using deep neural networks where features are learned through model training. By enhancing their flexibility and robustness, such systems can be easily applied in more educational scenes. At present, besides providing automatic scores in large-scale standardized testing contexts, automated writing assessment tools have been widely used in different educational contexts, and have transformed writing instruction in helpful and challenging ways (Palermo & Thomson, 2018; Wilson et al., 2021). In addition, when coupled with

tutorials or educational games, learning management features, or peer assessment, the tools could offer flexible, robust, and time-saving additions to the writing curriculum (Wilson & Roscoe, 2020).

2.2 Picture-cued writing and its application

Picture-cued writing tasks are widely used in building up and assessing language and writing skills. According to the Dual Coding Theory (DCT) theory, human beings use both visual and verbal representations to encode information in the cognitive system, and successful instruction ensures that learners build and integrate the two representations by using both visual and verbal cues (Paivio, 1991). By providing visual stimulation as well as meaningful situations, the use of pictures can also promote student motivation and engagement in language learning, and make students acquire a better understanding of language knowledge (Zhang & Zou, 2021). Aschawir (2014) postulates that the more interesting and varied the encouragement through visual aids, the faster and more effective the learning will be, contributing to learner interest and motivation. Asrifan (2015) similarly stated that “A visual material offers an attractive and stimulating framework for writing practice and has great potential as an aid to develop writing skills since it provides both contexts and stimulation of a variety of activities.” (p. 245).

Picture-cued writing is also widely used in language assessment as prompts to elicit written responses from students, especially those in the early stages of literacy education. Compared to textual writing prompts, picture-cued writing tasks are capable of presenting information in an intuitive and objective way, which are particularly suitable for young or beginning writers with low levels of literacy skills. Meanwhile, picture-cued writing can also provide meaningful context information to students, help them generate ideas, develop logical and critical thinking, and improve reasoning skills (Listyani, 2019).

Picture-cued writing tasks can be used to assess students at different levels of education. The internet Picture Naming Test (iPNT), for instance, is a web-based vocabulary assessment tool; it asks students to recognize and name the objects in pictures. iPNT is found to be a reliable measure particularly suitable for beginners and has a number of advantages over its paper-based and oral versions (Kharkhurin, 2012). Cross-linguistic Lexical Tasks (CLT) is another tool similar to iPNT. CLT is designed to assess receptive and expressive lexical skills for bilingual students through picture identification and naming of both nouns and verbs (Haman et al., 2015). Picture-cued writing tasks have also been applied to assess syntactic-level language learning, where students are required to write a sentence to describe a given picture (Silverman et al., 2015). More recently, picture-cued tasks are applied in higher education. Baird and Dooley (2017), for instance, proposed a visual post-entry language assessment approach to assess newly admitted university students, who were required to write a passage with visual prompts. The task was found especially suitable for students who have varying levels of comprehension and writing skills as well as different cultural and ethnic backgrounds, learning and formal English language instruction experiences. In short, picture-cued writing tasks have been widely used to assess student writing at the word, sentence, and passage levels.

2.3 The emergence of cross-modal learning

With the development of digital technology, various types of media data, such as texts, images, videos, and audios, have shown explosive growth in instructional activities, and instructors often use different kinds of data together to describe the same topic or tasks. For example, an abstract concept can be presented to students in an understandable fashion using visual images and scientific language. In this way, multimodal data are playing an important role in supporting instructional activities. Inherently, the human brain is a multimodal-multisensory dynamic learning system, outside information can be better perceived and understood by students with the help of multimodal data (James et al., 2017). For this purpose, how to efficiently process the data is very important, which can lead to a full application of them in education. Recently, deep neural networks have exhibited a powerful architecture to capture the relationship among the data in different models, and led to an important research direction: cross-modal learning (Wang, 2021). By analyzing and fusing data in multiple modalities simultaneously, cross-modal learning technology enables us to earn more comprehensive information about the concerned topic, and supports us to transform the way that some instructional tasks are performed, such as automatic assessment of the performance of the tasks involving multimodal data.

The purpose of cross-modal learning is to find the relationship between different modal data by using some deep learning models, and it covers any kind of deep learning that involves information obtained from more than one modality. Vision and language are two important media of human intelligence to understand the real world. Reasonably, a large amount of research has been done to bridge these two modalities in recent years (Lee et al., 2018; Malali & Keller, 2021; Wei et al., 2020). Among them, image-text matching is one of the fundamental topics in this field, which refers to measuring the visual-semantic similarity between a sentence and an image, and it has been widely adopted in various applications such as image captioning (Hossain et al., 2019), neural machine translation (Toyama et al., 2016), and especially the retrieval of text descriptions from image queries and image search for given sentences (Chen et al., 2020). In addition, people have been studying the matching and fusion techniques for data in other modalities. For example, Jin et al. (2021) introduced a method for matching audio and video data with a novel two-branch neural network, and Gabeur et al. (2020) presented a multi-modal transformer to jointly encode the different modalities in videos, which allows each of them to attend to the others. Benefiting from these emerging technologies, in this study, we aim to explore the feasibility of using these techniques to process multimodal data in open-ended writing situations, specifically trying to develop an automatic scoring framework for picture-cued writing tasks.

3 Methodology

3.1 Research design

A proper dataset is a foundation for developing an automated scoring model, as well as further answering our research questions. To the best of our knowledge, however, no dataset is available in the field of picture-cued writing currently. To this end, as

the pipeline in Fig. 1, we first created a dataset within two steps, 1) recruiting a large number of K12 students to take a picture-cued writing test, which included 15 picture description tasks, and collecting a set of picture-text pairs, i.e., textual responses and their corresponding visual pictures in the test; 2) grading the responses manually by human raters following a scoring rubric, and generating ground-truth scores for the responses. Then, we developed a novel automated scoring model by combining the emerging cross-modal matching technology and some classic NLP algorithms, which were used to measure the relevance between the student responses and the writing tasks, and evaluate the language quality of the responses, respectively. Finally, we evaluated the performance of the model and answered the research questions by comparing the machine scores, predicted by the model automatically, with the ground-truth scores at multiple dimensions.

3.2 Participants

In the current study, a total of 279 students in Grades 7 or 8 were finally recruited to participate in the picture-cued writing test, who were from 8 classes and 5 K12 schools in Mainland China. Students in Grades 7 or 8 were recruited based on two reasons. Firstly, they have been studying English as a second language (L2) for at least 5 years at school since the third grade of primary school, and most of them have the ability to write a complete sentence to describe a picture in English; Secondly, they are proficient in using a computer to complete the writing test conveniently, which is important for us to collect enough experimental data efficiently via a paperless online test.

For the participants, we also gathered some demographic information related to language learning and assessment, including their gender, age, grade level, and attitudes toward English learning. Among them, the grade levels were assigned by the students themselves through selecting one of three given options, i.e., top-1/3, middle-1/3, or lower-1/3, for a question of “What level do you think is your English grade in your class?”; and the attitudes were measured with 5 self-report scales from -2 to 2, which represent their attitudes toward English learning from very pleased

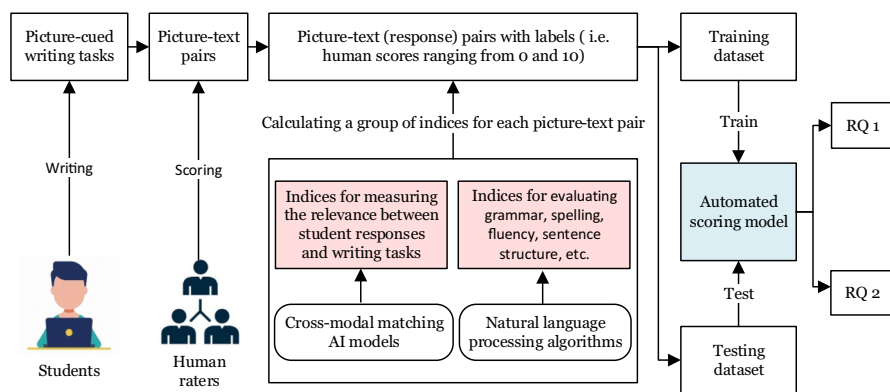


Fig. 1 The outline of the research pipeline

to very disgusted. The demographic variables were helpful for us to understand the validity of the writing test and evaluate the reliability of the automated scoring model.

3.3 Research materials

The writing test in the study was composed of a set of pictures, each of which was used to prompt students to write a sentence by applying their acquired vocabularies and sentence-level writing skills. Additionally, a writing instruction and a scoring rubric were also used in the study.

3.3.1 Pictures for writing

Overall, 15 pictures were selected and included in the writing task. Each picture was expected to take 1–2 min to be described by L2 students in Grades 7 or 8, and the entire writing test was scheduled to be completed in 40 min. Each of the pictures was selected carefully by two experts in the English teaching and assessment domain, and mainly considered the following two principles.

- (1) **A suitable level of vocabulary.** The participants were students in Grades 7 or 8 from k12 schools in China. For each selected picture, they should have learned enough vocabulary to describe it. To do this, the pictures were selected from the testing sets of COCO and Flick30, which are two of the most popular large-scale datasets available for public use, and each picture in the datasets has been provided with 5 ground-truth sentences written by skilled writers as good descriptions of the picture. The ground-truth sentences are helpful for us to select suitable pictures for the writing task. For each picture, taking the words appearing in its 5 ground-truth sentences as a reference, we make sure the participants have learned almost all the words or are capable of dealing with the very few unfamiliar words, for example using the familiar word “computer” to replace the proper noun “laptop”. This goal was achieved by checking whether each word in the ground-truth sentences appears in the vocabulary list of the English textbooks used in the schools.
- (2) **A variety of picture content.** Picture-cued writing is an effortful cognitive activity, participants first need to recognize the main objects, their status, and the context information in the picture, and then generate ideas and turn them into text by using vocabulary and grammar. This means that the variety of the image context affects the writing process and the quality of the writing. With consideration of the various contents of the pictures, we finally selected 15 pictures covering different themes, mainly including the natural world, popular sports, and daily life. Meanwhile, according to the number of salient objects and their interactions in the pictures, they can also be divided into three categories, and five pictures were available for each category. As the example pictures were shown in Fig. 2, each picture in the first category often contains one salient character (e.g., an elephant), each picture in the second category has two salient

characters (e.g., either two persons or one person plus one animal) as well as some interaction between the characters, and each picture at the third category contains multiple characters with a more complex context.

3.3.2 Writing instruction

To encourage students to describe the pictures comprehensively and submit high-quality responses as much as possible, a writing instruction was designed and presented to them at the beginning of the test as follows: *“Please observe each picture carefully, and write one sentence in English to describe the people, animals or things in the picture. Your responses should be focused on the questions: What is in the picture, what do they do, what they look like (how many, colors) and where are they. If there is a person or people in the photo, you will need to describe their appearance.”*

In order to support students from different schools taking the writing test conveniently and enable us to collect enough experimental data quickly, we further developed a questionnaire for the writing test by employing the online survey system Sojump (www.sojump.com). The questionnaire consists of three parts: 1) a sample demographic sheet for gathering participant information as the description in 3.2, 2) the above writing instruction, and 3) 15 picture-cued writing tasks. For each of the writing items, we displayed one of the selected pictures and provided a textbox below it (see Fig. 3(1)), and the textbox was used for students to type and submit their responses. By using the Sojump system, the time spent by each student to complete the writing task can also be automatically recorded.

3.3.3 Scoring rubric

The process of developing an effective automated scoring model always requires reliable ground-truth scores to train the model and evaluate its performance. For this purpose, a scoring rubric was designed by two domain experts, both of whom specialize in writing assessment research and are familiar with K12 English teaching. As listed in Appendix Table 5, the rubric consists of 6 components, i.e., grammar, spelling, convention, content comprehensiveness, content vividness, and sentence



(1) A picture in the first category (2) A picture in the second category (3) A picture in the third category

Fig. 2 Example pictures used in our writing test, which cover a variety of themes and are different in the level of content complexity. (1) A picture in the first category (2) A picture in the second category (3) A picture in the third category



(1) A part of the questionnaire for the writing test (2) Students were participating in the test

Fig. 3 Picture-cued writing test and data collection. (1) A part of the questionnaire for the writing test (2) Students were participating in the test

structure. Among them, comprehensiveness was mainly used to measure the extent to which students describe the content of the given pictures comprehensively, and other components were used to measure the writing quality in terms of linguistic characteristics. The total scores for six dimensions were 10 points, and clear criteria were also given for each component. Due to the importance of grammar and comprehensiveness on the writing test, both of them were scored on a three-point scale (i.e., score of 0 to 3), and other components were graded 0 or 1 point. The rubric was designed based on the fact that the test was conducted by asking students to write a sentence to describe a picture and their responses were mainly used to verify our research questions.

3.4 Data collection

With the designed questionnaire and scoring rubric, we collected experimental data and created a dataset for developing the automated scoring model through the following steps.

3.4.1 Conducting picture-cued writing

We conducted the picture-cued writing test in five K12 schools in China. During each test, one teacher from the schools was invited to organize the test, and students from each class were recruited to take the test simultaneously in a classroom. By providing a net-connected computer to each of the participants (see Fig. 3), they were asked to finish the test online within 40 min, and they were allowed to submit their responses early after they have described 15 pictures in the test or when they felt unwell. At the beginning of the test, the teacher was asked to introduce the writing instruction to the students. Furthermore, the participants were asked to complete the writing test independently, and they should not refer to the responses of their classmates or use any translation software. With the instructions

and requirements, we can collect more realistic responses, which are helpful for us to train and evaluate our automated scoring model reliably.

After two weeks of data collection, a total of 4185 responses were submitted by the 279 students for describing the 15 pictures in the writing task, and they spent an average of 29.6 min on the writing test. Among the data, the responses of 29 students are not suitable for training AI models as well as validating the research questions, which is because the responses contain Chinese characters, lack descriptions for some pictures, or be submitted with garbled texts. Thus, we removed the responses from our dataset, and the remaining 3750 responses of 250 students were used in the following section.

3.4.2 Human grading

Assigning a ground-truth score to each of the collected valid responses is the foundation for training and evaluating our model. For this purpose, four research assistants were selected as human raters to grade the responses manually according to the scoring rubric in Appendix Table 5. All of them have solid knowledge in the domain and prior experience with the English teaching and writing assessment, and the grading task was completed under the guidance of two experts who developed the scoring rubric as follows.

Stage 1: The two experts first trained the four assistants by explaining the scoring rubric and criteria, showing them how to grade the responses, and scoring 10 responses together. Through this round of grading, the assistants had a thorough understanding of the criteria.

Stage 2: Each assistant scored 95 responses independently, and their results were checked by the experts. Specifically, the responses scored differently by the assistants were highlighted, and the experts and assistants discussed the responses one by one until a consensus was reached.

Stage 3: The four assistants scored the remaining responses independently. Every two assistants worked as a pair. Pair 1 (Assistants A and B) graded the responses to pictures 1 to 8, and Pair 2 (Assistants C and D) graded the responses to pictures 9 to 15. As a result, each response was graded by two assistants.

Stage 4: After grading, each pair compared their scores and highlighted the different ones, and the assistants with two experts discussed these responses one by one and decided a final score for each of the responses together.

As a result, we obtained a dataset for training and evaluating our model, which contains a total of 3070 responses written by 250 students for 15 pictures as well as ground-truth scores assigned by human raters.

4 AI-assisted automated scoring

In writing assessment, automated scoring is often performed by evaluating student writings in terms of content achievement and language quality (Ramineni et al., 2013). The former is used to evaluate the extent to which a student responded to a

given writing topic relevantly and comprehensively, and the latter is used to evaluate the language quality in terms of grammar, spelling, lexis, structure, etc. In traditional writing assessments, both content achievement and language proficiency can be evaluated by using NLP algorithms, which is because writing prompts, student responses, and even reference answers are all in the form of a text. Differently, in picture-cued writing tasks, two modal data are involved, i.e., visual pictures and textual responses, and the content achievement of a response needs to be evaluated by judging its relevance to a given picture. To this end, we proposed to evaluate students' responses to picture-cued writing tasks by incorporating cross-modal matching technology and some NLP algorithms.

4.1 Estimating content achievement via cross-modal matching

For student responses to picture-cued writing tasks, the key to estimating their content achievements is to measure the similarities between them and their corresponding visual pictures. In recent years, the emergence of cross-modal matching deep learning allows us to match data across different modalities and estimate the similarities among them. For visual pictures and textual language, some research has been done to bridge these two modalities (Chen et al., 2022), and the visual-semantic similarity between them can be measured by using some image-text matching models by learning multimodal representations from large-scale image-text pairs. As shown in Fig. 4, image-text matching usually is achieved with three encoders, i.e., an image encoder, a text encoder, and a multimodal encoder. Given an image and a text, the image encoder is often used to detect salient regions from the image and generate a visual representation for each region, the text encoder is used to parse the text and generated a contextualized textual representation for each word in the text, and the multimodal encoder is used lastly to estimate a visual-semantic similarity between the image and the text by matching the visual and the textual representations. How to select the three components has a direct impact on the performance of image-text matching. By considering the diversity of student responses and the complex content in each picture, we utilize three image-text matching methods, namely Graph Structured Network for Image-Text Matching (GSMN) (Liu et al., 2020), Similarity Reasoning and Filtration for Image-Text Matching (SGRAF) (Diao et al., 2021), and ALign Image-Text Representations BEfore Fusing (ALBEF) (Li et al., 2021), to estimate the content achievement in our automated scoring method. These three methods are typical in the selection of the encoders, and have been highlighted in multiple vision-and-language tasks recently.

Both GSMN and SGRAF use Faster-RCNN (Region-based Convolutional Neural Network) as an image encoder and use a Bi-directional Gated Recurrent Unit Neural Network (BiGRU) as a text encoder. Faster R-CNN is a widely used network pre-trained for detecting salient regions and identifying objects in an image based on deep learning (Ren et al., 2016), and BiGRU is a classic semantic analysis tool known for extracting contextual representations through analyzing a text in both forward and backward orders (Schuster & Paliwal, 1997). The difference between GSMN and SGRAF is the architecture of the multimodal encoder. By explicitly

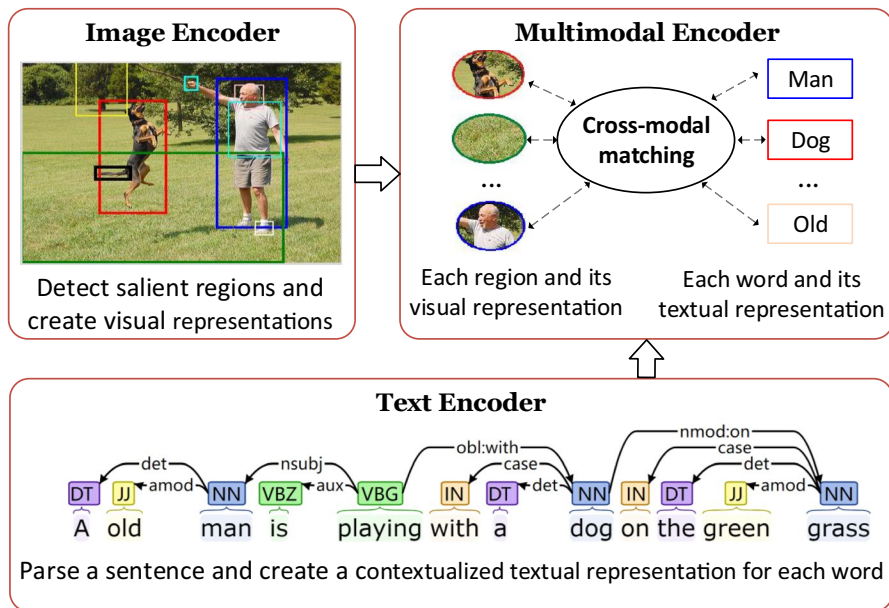


Fig. 4 A typical framework for cross-modal matching

modeling objects, relations and attributes as structured phrases, e.g., “an old man” or “playing with a dog”, the multimodal encoder in GSMN is capable of learning fine-grained correspondences between an image and a sentence at word-level matching and phrase-level matching. With an introduction of a similarity attention filtration, the multimodal encoder in SGRAF can capture more comprehensive interactions between the visual and textual modalities by stressing the key and significant alignments (e.g., “man”, “old” and “dog” for the image and the sentence in Fig. 4) and meanwhile casting aside the interferences of irrelevant alignments (e.g., “is”, “with”, “of”, and etc.).

Distinctly different from GSMN and SGRAF, ALBEF is totally based on the Transformer architecture (Vaswani et al., 2017), which has demonstrated powerful ability in natural language processing and promising potentiality in visual tasks (Tan & Bansal, 2019). As a novel pure attention-only sequence-to-sequence architecture, the Transformer is beneficial for ALBEF to better understand the textual and visual representations. In addition, thanks to a self-training method that learns from pseudo-targets produced by a momentum model, the multimodal encoder in ALBEF is able to deal with varied sentences written for the same picture more robustly. For instance, the picture in Fig. 4 can be described as “The old man is playing with a dog on the lawn.” or “An old man is training his dog on the grass.” Although they are different in writing, they both describe the content of the image well and the method tends to output close matching results for them. Because student responses are always different, this characteristic is especially meaningful for writing assessments.

In short, each of the three typical image-text matching methods has its own advantages. For this purpose, we employed them simultaneously in this paper

to estimate a group of similarities between a visual picture and a textual student response, and the similarities were considered as indices to evaluate the content achievement of the response, i.e., measure how well the response describes the content of the picture.

4.2 Evaluating language quality via natural language processing

Besides content, language quality is also important in writing assessments. Intuitively, the writings with more grammar errors or a lack of fluency should be assigned with lower scores. In this paper, the language quality was evaluated by using some NLP techniques as well as deep learning-based language models. Specifically, we first analyzed a student response by using Stanford CoreNLP (Manning et al, 2014), including tokenizing it into words, assigning a part-of-speech tag to each word, and extracting a dependency parse tree for it. On the basis of the analysis, we generated a group of indices related to vocabulary use, syntactic dependencies, sentence structure, etc. Meanwhile, several deep learning-based language models were also employed to evaluate the writing quality of the response. The models were usually pre-trained on a massive corpus, and have a powerful ability for grammar check as well as extracting some indices related to the fluency and conventions of a writing text. Considering the student responses in our writing test to mainly be a single sentence, we mainly used the following categories of indicators to measure their language quality.

- (1) A set of word-count-based indices, n_{word} , n_{con} , n_{adj} , n_{adv} , n_{wh} and n_{pp} , which represent the number of words, conjunction, adjectives, adverbs, pronoun and prepositional phrases, respectively. Among them, n_{word} is commonly considered to be related to writing ability, and it has been validated as a valuable indicator for automated writing assessments (Liu et al., 2019); n_{adj} , n_{con} , n_{adv} and n_{pp} can be used to measure the degree to which students describe a picture in detail and vividly.
- (2) The number of grammar errors and spelling errors, denoted as n_{s_error} and n_{gm_errors} . At present, many powerful tools are available for us to perform spelling and grammar checks. In this study, a popular writing assistant, Grammarly (<https://www.grammarly.com>), was employed to count n_{s_error} and n_{gm_errors} , and the typical errors include incorrect subject-verb agreement, wrong tense or verb form, mistakes in the use of articles, pronoun errors, and incorrect word form.
- (3) A perplexity score, $ppl(S)$, where $ppl(S) = \exp\left\{-\frac{1}{n} \sum_{i=1}^n \log p_{\theta}(x_i)\right\}$ was estimated using GPT2. As one of the most famous language models, GPT2 was defined as a probability distribution over sequences of words, and can predict the next word in a sequence given the prior text. Based on the characteristic, we used the model to compute a perplexity score. As defined in formula (1), where S is a sentence, n is the number of words and $p_{\theta}(x_i)$ is the probability that the i^{th} word appears in the sentence. A high score will be obtained when a sentence is not likely to occur commonly, lacks fluency, or contains some grammar or spelling errors.

- (4) Two frequency-related indices, f_{word} and f_{bigram} , which are the mean of log frequencies of all content words and the mean of log frequencies of any two adjacent words (i.e., a bigram) in a sentence. The frequency is calculated according to the number of times a word or a bigram appears in a corpus. In our study, the COCA corpus was used. Obviously, f_{word} is highly related to the student's vocabulary, and a low f_{word} will be generated if a student writes a sentence only using some simple and common words; f_{bigram} is useful for measuring the fluency and conventions of a sentence.
- (5) The number of clauses in a response, n_{clause} . In our writing test, we asked students to write a sentence to describe a given picture, and encouraged them to describe the picture in as detail as possible. As a result, students with high English writing skills tend to submit a compound or complex sentence. Based on the extraction of a dependency parse tree for a sentence, we obtained n_{clause} by counting the number of clauses in the sentence.

4.3 Automated scoring model

As mentioned above, we collected a set of student responses to picture-cued writing tasks, and each response was graded manually by human raters and a group of indices was estimated for it to measure its content achievement and language quality. Based on the dataset, we trained an automated scoring model in this section. The 75% of responses in the dataset were placed in our training set and used to build the automated scoring model, and the remaining 25% were placed in our testing set and used to evaluate the performance of the model. Because the performance of the model may vary based on how the data were divided into the training and testing sets, we used a cross-validation technique across four separate holdout sets to ensure the generalizability of the model. In this way, we can observe whether the performance of the model is sensitive to particular features of the training and testing sets, and understand how our results are likely to generalize to new data.

With the training sets, we trained our automated scoring model using the supervised machine learning paradigm. To improve the accuracy and robustness of the model, three strategies were used in our training process. At first, considering the scores of the responses are imbalanced, especially there are only a few responses assigned with a low score (e.g., only 0.4% and 0.61% responses scored 0 and 1) and a high score (e.g., only 0.53% responses scored 10), some over-sampling techniques were applied to the training set for alleviating the impact of data imbalance, including the naive random over-sampling and the synthetic minority oversampling (SMOTE). The former randomly repeats some samples from the minority classes and balances the training sets, and the latter analyzes the samples from minority classes and synthesizes some new samples artificially for the dataset (Chawla et al., 2002). Secondly, some feature selection techniques were also considered on the training set, such as Recursive Feature Elimination (RFE), which recursively removes the features with the least importance score, and model-based variable selection methods, which select features by linear models penalized with the L1 norm. Thirdly, multiple popular machine learning models were considered,

including Random Forest, Support-vector Machines, and XGBoost. Each model was trained on the training set, and a grid-search technique was applied for hyper-parameters tuning, details please refer to Appendix C.

After training the models on different datasets repeatedly, the XGBoost model was finally selected as our automated scoring model due to its high accuracy and good stability. The basic idea behind this algorithm is to build a weak model that draws conclusions about various feature importance, and then build a new, stronger model using these conclusions, finally it will predict the sentence scores automatically. The used indices as well as their importance were listed in Appendix D.

5 Results and discussion

After the model was trained, we first used it to predict machine scores automatically for the responses in the testing sets, and then evaluated its performance by investigating the agreement between the machine scores with the corresponding human scores.

5.1 Automated scoring and evaluation metrics

Six metrics were used in our evaluation (see Table 1), which are standard and popular in the field of machine scoring. Among them, MAE , A_{exa} and A_{adj} are usually used to explicitly measure the accuracy of machine scores, and QWK , r and SMD are used to evaluate the reliability and usability. As the widely used criteria proposed by Williamson et al. (2012), QWK and r between machine scores and human scores must be at least 0.7, indicating that at least half of the variation in human scores is accounted for by machine scores; meanwhile, SMD between machine scores and human scores cannot exceed 0.15. These criteria were used in the following evaluation of this study.

5.2 Result for the first research question

Our first research question concerns how effectively the cross-modal matching AI technology can be used to measure the degree of a picture being described comprehensively by students in open-ended writing situations. For this purpose, the component named *content comprehensiveness* was involved in our scoring rubric (see Appendix Table 5), which requires assigning a sub-score ranging from 0 to 3 to measure the matching degree of a picture and a response written to describe it. For each student response collected in our study, a human sub-score has been assigned by manually comparing the relevance of its contents with the corresponding picture. Meanwhile, for the automated scoring of the content comprehensiveness, our method has estimated a group of similarity indices using the three cross-modal matching AI models, i.e., GMSN, SGRAF, and ALBEF, and predicted a machine sub-score based on the similarity indices automatically. Here, we answered the first

Table 1 Metrics used to evaluate the performance of our method

Metrics	Abbr	Description
Mean absolute error	MAE	The average of the absolute differences between pairs of machine and human scores, and a lower value indicates machine scores are closer to human scores with a better performance
Exact-agreement percentage	A_{exa}	The percentage of student writings being scored by human raters and automatic tools with an absolute agreement, i.e., human and machine scores match exactly
Adjacent-agreement percentage	A_{adj}	The percentage of student writings being scored with an acceptable agreement, i.e., the absolute difference between human and machine scores is less than a threshold (1 in this paper)
Quadratic weighted kappa	QWK	QWK provides a measure of agreement beyond chance, and can obtain a better reliability coefficient on imbalanced data sets. A perfect score of 1.0 denotes a complete reliability between human raters and machine tools whereas a score of 0 means a random reliability
Pearson correlation	r	r is defined in statistics to measure how strong a relationship is between human and machine scores, and it assigns a value between -1 and 1 , where 0 is no correlation, 1 is total positive correlation, and -1 is total negative correlation
Standardized Mean difference	SMD	SMD is the difference between the means of human and machine scores in standard deviation units, it is used to evaluate whether the distribution of scores from automated scoring is centered on a point close to what is observed in human scores

research question by analyzing the relationship between the AI-generated results and the human-scored results.

In Fig. 5, for student responses to our 15 pictures, we first calculated the Pearson correlation r between their human subscores and the original similarity indices estimated by the cross-modal matching models. The results derived from GSMN, SGRAF, and ALBEF were represented by the points on the red, green, and blue lines, respectively. The results show that r is between 0.3 to 0.7 in most cases, indicating that the indices are moderately to highly correlated with the human subscores. In other words, the cross-modal models can generate useful indices for the measurement of the matching degree between the visual pictures and their textual responses. Meanwhile, the results also reveal that r derived from one of the models is not good enough and unstable across different pictures. By integrating the results of the three models, we further calculated r using the mean of all similarity indices (see the dotted black line) and the machine subscores predicted based on the indices (see the solid black line). With the combination of the models, we found that r was significantly improved and achieved a strong correlation (i.e., $r > 0.7$) in most cases, and presented better stability only varying slightly in the range of 0.7 to 0.9 across the 15 pictures. It is the reason that we integrated the cross-modal models into our method.

Furthermore, we investigated the agreement between the machine and human sub-scores assigned for the component of content comprehensiveness by calculating A_{exa} and MAE . On our four testing sets, the average A_{exa} and MAE achieved 80.7% and 0.19, respectively. The results mean that 80.7% of the student responses were scored by our automated method and human raters with an absolute agreement, and only 0.19 difference exists between the human and machine sub-scores for all the responses. These results show that our method can measure the matching

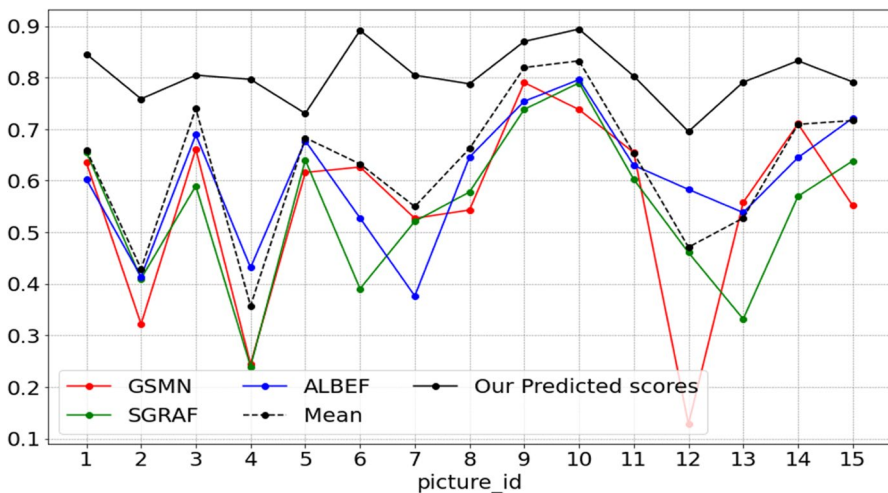


Fig. 5 Pearson correlation r between human sub-scores, assigned for the content comprehensiveness, and the similarity indices estimated using GSMN, SGRAF and ALBEF, and the mean of the similarity indices and the machine scores predicted by our method

degree between a visual picture and a textual description effectively, and estimate the content achievement in the picture-cued writing as human raters. In addition, we presented an example for understanding the picture-text matching intuitively in Fig. 6. For the response of “A father was patiently teaching his child to ride a bicycle on the grass”, which was written by a student for the 5th picture in our test, Fig. 6 illustrates the heatmaps on the picture corresponding six key words in the response. From the example, we can find that the AI model has matched visual areas to the target words reasonably after a cross-modal deep learning.

5.3 Result for the second research question

Our second research question mainly asks: whether it is possible to develop an automated scoring method for picture-prompt writing tasks by combining NLP and cross-modal deep learning? To answer the question, with all indices estimated by the cross-modal matching models and NLP algorithms, we predicted a machine total score ranging from 0 to 10 for each response in the testing sets by using our method, and investigated the agreement between the machine total scores and the total scores graded by human raters.

5.3.1 Overall accuracy of automated scores

As listed in Table 2, on all the testing sets, our method obtained small *MAE* and *SMD* as well as high enough A_{exa} , A_{adj} , *QWK*, and *r*. Among them, *MAE* was less than 0.5 on all the testing sets, and its average achieved 0.479, i.e., the average difference between the machine and human scores was 0.479. For the total score of 10, we considered that a difference of approximately 0.5 was acceptable. In terms of A_{exa} and A_{adj} , their lowest values of 63.76% and 90.61% were obtained both on the third testing set, indicating that our method determined at least 63.76% of student responses accurately and 90.61% only with a bias of no more than 1. In addition, both *QWK* and *r* exceeded 0.85 and *SMD* was less than 0.1, which also met the widely accepted criteria that both *QWK* and *r* must be at least 0.70 and *SMD* cannot exceed 0.15. In Fig. 7a and b, we further displayed the distributions of the human and machine scores by counting the number of student responses at each score from 0 to 10. Obviously, the distributions of the human

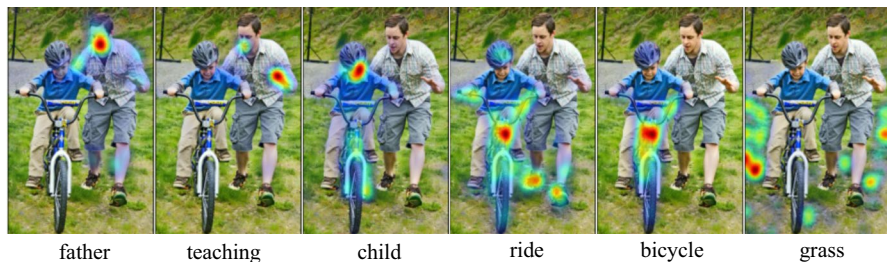


Fig. 6 The cross-attention heatmaps corresponding to the key words in the response of “A father was patiently teaching his child to ride a bicycle on the grass”

Table 2 The agreement between human and machine scores

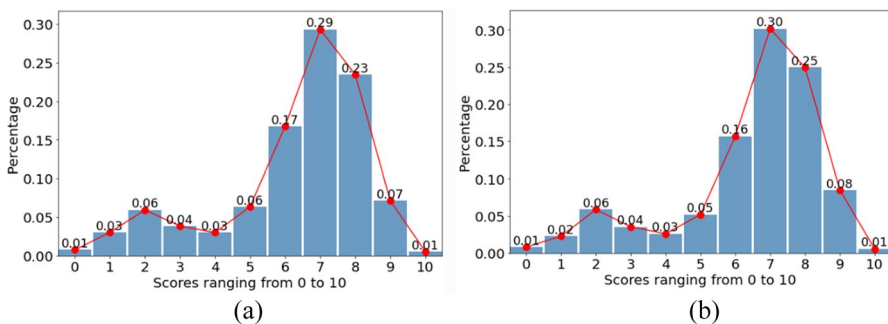
Fold	N	MAE	A_{exa}	A_{adj}	QWK	R	SMD
1	938	0.474	63.64	91.36	0.913	0.914	0.052
2	938	0.475	63.86	90.72	0.914	0.916	0.078
3	937	0.491	63.39	89.86	0.908	0.910	0.073
4	937	0.475	64.14	90.50	0.912	0.915	0.080
Average	0.479	63.76	90.49	0.912	0.914	0.071	

The row marked average provides the average statistics across the 4 hold-out testing sets

and machine scores are very similar, i.e. the percentages of the responses with the same scores are also close in the two distributions. All the results indicate that our method has scored student responses in the picture-cued writing tasks in a good agreement with human raters. Meanwhile, all the six metrics obtained on the different testing sets are very similar, which means that our method can generalize well on the datasets.

5.3.2 Adaptability to different pictures

As shown in Appendix A, three categories of pictures were involved in our writing tasks, and their complexity increased gradually due to an increase in the number of characters and objects in them. To investigate the adaptability of our scoring method for the different pictures, we calculated MAE , A_{exa} , and r for the three categories of pictures, respectively. In line with expectations, more complex pictures actually posed greater challenges to our automated scoring method. As shown in Fig. 8, with the rise of the complexity in the three categories, MAE between human scores and machine scores increased gradually, and both A_{exa} and r decreased gradually. Nevertheless, the degradation in performance is slow and the scoring results for all pictures are acceptable. Even on the most complex pictures in the third category, our model scored 61% of student responses exactly as human raters and only yielded an error of 0.52 for the 10-point writing test.

**Fig. 7** Distribution of human scores (a) and machine scores (b)

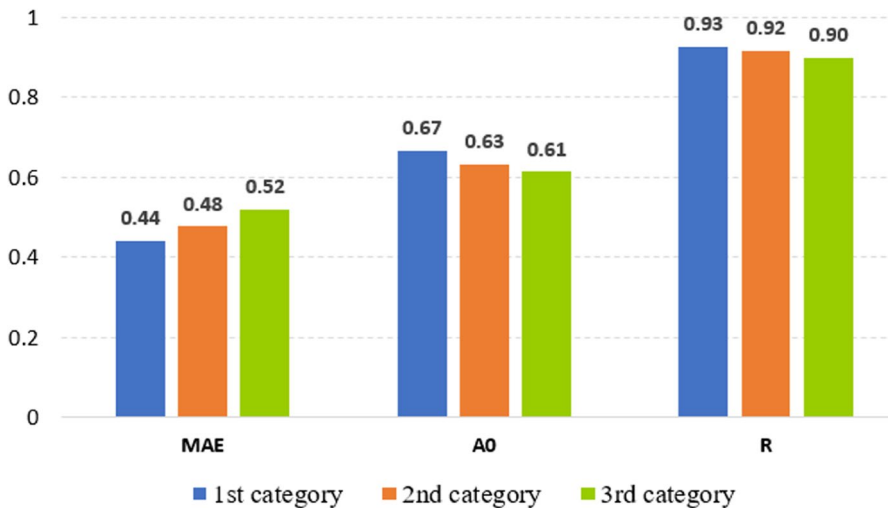


Fig.8 Statistics of the scoring results for three categories of pictures with different levels of complexity

5.3.3 Fairness across subgroups

Although the human and machine scores are highly related on the whole population, this does not eliminate the possibility of a significant difference existing between them for some subgroups of interest (Bridgeman et al., 2012). In this research, we further explored whether our method has the ability to detect subgroup differences in writing performance. According to the participants' background information, which were collected synchronously during the writing test, they were divided into subgroups in terms of their gender, grade level, and attitudes toward English learning. For each subgroup, we calculated the mean and standard deviation of the human scores and machine scores separately, as well as the standardized mean difference (SMD). As the results listed in Table 3, the machine scores and human scores have similar means and standard deviations for each subgroup, and all SMD are less than 0.1. The results indicated that our method scored student responses in different subgroups as fairly as humans do, and predicted similar results as humans. In addition, by comparing the mean scores in the subgroups, we can find that the mean scores derived from our method are varied rationally across the subgroups and have a consistent trend with human results. For example, in the terms of the attitude toward English learning, students were categorized into five groups according to their attitudes from negative to positive, the mean scores increase gradually from 4.689 to 6.626 as the attitude changes from most negative to most positive. The same conclusion can also be observed from the subgroups of gender and grade levels.

6 Discussion

This study aimed to develop an automatic scoring method for picture-cued tasks. For the purpose, we first designed a picture-cued writing test of 15 tasks and collected 3750 valid responses from 250 k12 students in China, and created

Table 3 Human–Machine agreement for subgroups in terms of participant’s gender, grade levels, and their attitude towards English learning

Category	Subgroups	N	H_Mean	M_Mean	H_SD	M_SD	SMD
Gender	Female	1905	6.296	6.455	2.042	1.997	0.077
	Male	1845	6.358	6.492	2.128	2.097	0.065
Self-evaluation grade levels	Top 30%	1516	6.739	6.873	1.822	1.791	0.065
	Middle	1529	6.401	6.536	1.926	1.856	0.066
	Bottom 30%	705	5.275	5.475	2.542	2.562	0.097
Attitude towards English learning	 Positive 2 1 0 -1 Negative -2	1020	6.626	6.762	1.796	1.747	0.066
		1214	6.477	6.623	1.999	1.934	0.071
		1216	6.165	6.325	2.106	2.083	0.078
		210	5.638	5.748	2.745	2.727	0.053
		90	4.689	4.878	2.787	2.867	0.091
		-1					

N represented the number of student responses in each subgroup, H and M represented human scores and machine scores, Mean and SD were the mean and standard deviation of the scores, and SMD was the standardized mean difference between human and machine scores

a dataset for training and evaluating our AI model. Then, a group of feature indices was estimated by using the emerging cross-modal matching technology and some NLP algorithms. These indices allowed us to measure the similarity between a visual picture and a textual description, and to evaluate the language quality of the description in terms of grammar, spelling, fluency, structure, etc. Finally, an automated scoring model was developed by using a supervised machine learning classifier, and its performance was assessed by investigating the agreement between human raters and the AI model. The assessment showed that our model can grade student responses for picture-cued tasks as fairly as human raters.

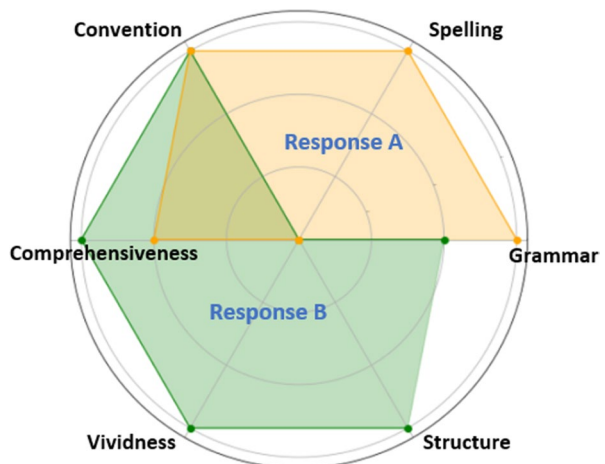
For an intuitive understanding, we presented a group of examples for our method in Table 4. For the left picture, student responses were automatically graded into four levels by our method according to their similarity to the picture. For each level, we listed three example responses on the right. All the responses graded at the highest level with 3 points described the content of the picture comprehensively, including the main subjects (two persons), their actions (learning to ride a bicycle), and the context (on the grass) in the picture. The responses graded into the second level with 2 points, except for the absence of the context, provided good descriptions including the main subjects and their action. In contrast, the responses graded into the third level with 1 point only included a small part of the picture content. Taking the response of “The boy is learning how to ride a bike” as an example, it lacked the description of another subject and the context. The responses, which were completely irrelevant to the content of the picture or too simple, were graded at the lowest level with 0. These examples further demonstrated that our method can estimate the content achievement for picture-cued writing tasks by measuring the matching degree between a visual picture and a textual description.

Table 4 A picture and some example responses for four scoring levels from 0 to 3.

A picture for writing	AI-Scores	Example responses at four scoring levels from 0 to 3
	3	A father was patiently teaching his child to ride a bicycle on the grass. On the grass, boy is trying to keep his balance on the bike, his father is following him to ensure his safe. A middle-aged man is teaching his child to ride a bike in the grassland.
	2	A children is trying to ride the bicycle with the help of his brother. The elder people is teaching the kid to ride a bike. The young boy is learning to ride a bike and his father is protecting him.
	1	The boy's farther are teaching his child. A young child is struggling to learn to ride a bicycle. The boy is learning how to ride a bike.
	0	The father. I am sorry, I can not boy is

Additionally, because our method scores student writings on six dimensions, including grammar, spelling, convention, comprehensiveness, vividness, and structure, it is feasible to provide fine-grained feedback to students for improving their writing. Figure 9 is an example of such diagnostic feedback; the figure presents six sub-scores generated by our method for two student responses to Picture 5 in Appendix A. Response A is “The boy is learning how to ride a bike”, and Response B is “On the grass, boy is trying to keep his balance on the bike, and his father is following him to ensure his safe”. Response A obtained lower scores in terms of comprehensiveness, vividness, and structure because it only mentions one character in the picture and does not use any adjectives and adverbs to describe the properties and actions of the character. As feedback, we

Fig. 9 Assessment results for two responses (Response A is “The boy is learning how to ride a bike”, and Response B is “On the grass, boy is trying to keep his balance on the bike, and his father is following him to ensure his safe”) from their grammar, spelling, convention, comprehensiveness, vividness, and structure



can encourage the student author of Response A to describe the picture more accurately and comprehensively. Using modifiers of nouns (adjectives) and verbs (adverbs) could enhance description accuracy, whereas adopting a compound sentence structure could effectively combine information about multiple characters. To the author of Response B, which is a more comprehensive description of the picture though it contains some spelling and grammar errors, the feedback could focus on strategies to improve linguistic accuracy.

In our experiments, discrepancies were also found between the AI model and human raters in some cases. For instance, compared to simple pictures with less salient objects and a pure background (such as Fig. 10c), complex images (such as Fig. 10a and b), which typically include a group of people, animals, or various objects in a more complex environment, tend to yield a larger *MAE* and lower correlation r (see Fig. 8). The disagreement between machine and human scores seems to increase as the complexity of the image increases. The main reason is that the AI model is trained to reward descriptions with more objects in the picture while human raters usually focus more on the few salient objects as well as the language quality of the writings. Taking Fig. 10b as an example, human raters assessed the salient objects in the picture, including children, birthday cake, and table, according to the scoring rubric. They assigned the highest score of 3 to the responses such as “A group of children and some adults gathered around the table to watch the colorful cake” and “A group of adults and children gathered around the dinner table for a child’s birthday”. The AI model, however, considers more elements in the picture and rewards the inclusion of details. For instance, the model assigns a high score to responses such as “Surrounded by men and women as it had the apples on a table mat for the boys to celebrate the birthday” and “It is a birthday, but the cake on the left is simpler than the cake on the right”, whereas the same responses receive lower scores assigned from human raters, probably because of their inadequate focus on the salient objects. A similar discrepancy was also observed for the 12th picture in Fig. 10a. Responses such as “Two students raise their hand and the boy looks at the girl” and “A boy is competing with a girl” receive scores from our modal that are higher than the scores given by human raters.

Another noteworthy observation is that the naming of the objects may affect the scoring results. For example, the 5th picture presents a scene



Fig. 10 Example pictures with some typical discrepancies between human scores and machine

where a woman wearing sunglasses sits on a red bench (see Fig. 10c). Compared with other pictures used in the writing test, this picture is simple as it contains one salient character (i.e., the woman) and a few other less-salient objects. Therefore, a more accurate scoring result was expected from the model. However, the picture has a low correlation r and a large MAE. A close analysis of the responses found this is due to the naming of the objects, specifically, the red bench was referred to as a chair by many students, and the model assigned lower scores to the responses containing “chair” than those containing “bench”. This interesting instance indicates the students’ limited vocabulary size at this stage of their English learning. They have learned general words (a hypernym, such as ‘chair’) but not yet the specific ones (a hyponym, such as ‘bench, couch, sofa’) to describe an object more accurately.

7 Conclusion

In this paper, we proposed an automated method for scoring student responses to picture-cued tasks by incorporating the emerging cross-modal matching technology and some NLP algorithms. The findings from this study provided preliminary validity evidence to support the use of the method in language instruction, which not only can reduce teachers’ grading workload but also provide learners with timely feedback. Furthermore, since multimodal data have been widely used in education, this study provides some inspiration for handling similar tasks by using the emerging cross-modal AI technology.

One limitation of the present study is associated with the relatively small sample size. Specifically, we had insufficient student responses scored at the low and the high ends, that is, only 20 responses scored as zero at the lowest and 28 marked as 10 at the highest. Because the automated scoring model needs to map the extracted linguistic features and the estimated similarities to different score levels during the training of the model, the limited size of the training dataset restricts the model to identify all the unique features associated with each score. Sakaguchi et al. (2015) investigated the impact of training data characteristics on automated short-answer scoring models, and they found that the number of examples per score level is associated with the performance of automated scoring models.

Moving forward, we will create a large-scale dataset by recruiting students to describe more pictures, including pictures taken by themselves in authentic environments, and improve the adaptability of the automated scoring method and enable it to handle various responses to picture-cued writing tasks reliably. In addition, we intend to invite teachers to use our automated scoring model in their instruction, and investigate the agreement between our model and human teachers and explore how effectively our model facilitates language teaching and learning in practice.

Appendix 1

All 15 pictures were used in our writing test, and they were grouped in three categories. Each picture of first category includes one dominant character, e.g., a person or an animal, each picture of second category covers two characters having some interaction, and each picture third category often includes a group of people in an activates.

Pictures of first category	Pictures of second category	Pictures of third category
 (1)	 (2)	 (3)
 (4)	 (5)	 (6)
 (7)	 (8)	 (9)
 (10)	 (11)	 (12)
 (13)	 (14)	 (15)

Note. The numbers are the ids of the pictures in our research and this paper.

Appendix 2

Table 5 The scoring rubric designed for the picture-cued writing test in this study, and all the example responses were written by students for the 8th picture in Appendix A

Components	Levels and scoring criteria	Scores	Two example responses for each level
Grammar ∈ [0, 3]	There are serious grammar errors, and the response is not comprehensible	0	<i>A old man play with old dog people</i>
	There are some grammar errors, but the response is comprehensible	1	<i>A old man play with a dog A man and a dog</i>
	There are very minor grammar errors	2	<i>The old is teaching the dog how to jumping</i> <i>The old man is play with his dog</i>
Spelling ∈ [0, 1]	The grammar of the response is perfect	3	<i>An old man is playing with a dog</i> <i>An old man is training his dog with a stick</i> <i>An old man is waiking a dog</i>
	There are spelling errors	0	<i>A man is palying with a dog with a branch</i> <i>A man plays a dog with a stick on the lawn</i> <i>Play with a dog</i>
	The spelling is perfect	1	<i>The man is practice the dog's jump</i> <i>Thers are adog and aold man</i> <i>A dog plays with its owner</i> <i>He is playing with his dog</i> <i>I am sorry, I can not a dog</i> <i>A dog jumps highly</i> <i>A person and a dog</i>
Convention ∈ [0, 1]	There are space or punctuation errors	0	<i>The old man is playing with his dog</i> <i>The man is teasing his dog with a branch</i>
	The space and punctuation are perfect	1	<i>An old man was teasing the dog with a stick on the lawn</i> <i>An old man is using a bough to play with a black and yellow dog on grass in the park</i>
	There are space or punctuation errors	2	
Content comprehensiveness ∈ [0, 3]	The response content is completely irrelevant to the picture or too simple	0	
	The response content includes only a small part of the image content	1	
	The response content includes a big portion of the image content	2	
	The response covers three aspects of the picture, i.e., the main subjects, their actions or status, and the context, one score for each. If there are two or more subjects, while the response only covers one, it will not get 3 scores	3	

Table 5 (continued)

Components	Levels and scoring criteria	Scores	Two example responses for each level
Content vividness $\in [0, 1]$	There is no descriptive detail at all	0	<i>A man is playing with a dog</i> <i>The dog playing with its master</i>
	There are descriptive details	1	<i>The black dog jumps high to catch the trick</i> <i>A man with gray hair is playing outdoors with his dog</i>
Sentence structure $\in [0, 1]$	It is a simple sentence	0	<i>An old person is playing with the black dog</i> <i>On the grass, an old man is teasing his dog</i>
	It is a compound or complex sentence	1	<i>An old man is using a stick to play with the dog, whose face is full of smiles</i> <i>The dog jumps because his master commands him</i>

Appendix 3. Machine learning model training.

Considering the problem of imbalanced data, that there are only few responses assigned with a low and high score. Serious imbalance of data can even cause the trained model to directly ignore categories with a small sample size. Some optimization tricks can be used to make the model pay more attention to the classes with fewer samples. This allows the model to focus on learning such features, rather than focusing too much on categories with a large number of samples. A simple way is to copy the same data directly to achieve the purpose of expanding the data, that is, oversampling. Another method is SMOTE. First, each sample x_i is selected from the minority class samples as the root sample for synthesizing new samples; secondly, the k nearest neighbor samples of the same class of x_i are used as the reference for synthesizing new samples, and new samples are generated by interpolation, repeat this process until the required number of samples is reached. Here, k is generally an odd number, we have tried $k = 5, 15$, and 25 on the training set, and found that $k = 5$ works better. Two sampling methods are compared, and finally we choose oversampling as the method for balancing the data based on their performance on the training set.

Next, we trained several machine learning models and used grid search to tune the parameters of each model by tenfold cross-validation on the training set. These models are: K-nearest neighbors algorithm (k-NN) (Coomans & Massart, 1982), which classify samples by finding the k most similar, nearest neighbors. The searching space of hyperparameters is {Number of neighbors: 5, 10, 20, 50. Weight function: “uniform”, “distance”}. The second model we trained is Random Forest (Breiman, 2001), which builds multiple decision trees in order to combine all predictions for a more robust behavior. The searching space of hyperparameters is {Number of decision trees in the forest: 50, 100, 200, 400. Criterion: “gini”, “entropy”. Maximum depth of the tree: 4,5,6,8,10. Minimum number of samples to split: 20,40,80,100. Number of features to consider when split: “sqrt”, “log2”}. Support-vector machines (SVM) (Cortes & Vladimir, 1995) is also considered, it maps training samples to point in space for classification and regression analysis. The searching space of hyperparameters is {Regularization parameter: 1, 10, 100. Kernel: “linear”, “poly”, “rbf”, “sigmoid”. Kernel coefficient gamma: $1e-2$, $1e-3$, $1e-4$, “auto”}. Another model we trained is XGBoost (Chen & Guestrin, 2016), which implements machine learning algorithms under the Gradient Boosting framework. The searching space of hyperparameters is {Learning rate: 0.05, 0.1, 0.2, 0.3. Maximum depth: 4, 6, 8. Subsample: 0.6, 0.8, 0.9, 1. Scale_pos_weight: 1, 5, 10. Alpha: 0, 1, 2, 5, 10}. Based on the model performance on the tenfold CV, the models with the least MAE were selected for each component. We trained models to predict 6 components first, and the performance are list in the following table. By adding scores of these components, we get the final scores. Compared with the human scores, the total MAE is 0.479.

Components	MAE
Grammar $\in [0, 3]$	0.230
Spelling $\in [0, 1]$	0.056
Convention $\in [0, 1]$	0.054
Comprehensiveness $\in [0, 3]$	0.198
Vividness $\in [0, 1]$	0.083
Sentence structure $\in [0, 1]$	0.018

Appendix 4

Main indices used in our automated scoring model. In the figure, the values represent the Gini importance (Steinberg & Colla, 2009) of the indices in evaluating the grammar, spelling, convention, comprehensiveness, vividness, and sentence structure of student response, as well as in predicting the final score for the responses. The larger the values, the more important the indices are.

Variables	ALBEF_1	0.10	0.00	0.12	0.31	0.00	0.14
	ALBEF_2	0.01	0.00	0.00	0.08	0.00	0.00
	GSMN_1	0.00	0.01	0.00	0.14	0.00	0.00
	GSMN_2	0.01	0.01	0.00	0.10	0.01	0.00
	SGRAF_1	0.01	0.00	0.03	0.10	0.01	0.02
	SGRAF_2	0.00	0.00	0.00	0.15	0.06	0.00
	f_words	0.02	0.02	0.01	0.01	0.23	0.02
	n_adj	0.05	0.05	0.00	0.00	0.35	0.00
	n_adv	0.00	0.12	0.00	0.00	0.22	0.00
	n_clauses	0.06	0.00	0.06	0.01	0.00	0.31
	n_con	0.19	0.00	0.35	0.01	0.00	0.00
	n_grammar	0.15	0.02	0.01	0.02	0.03	0.01
	n_spelling	0.00	0.60	0.00	0.00	0.03	0.00
	n_upper	0.00	0.00	0.00	0.00	0.00	0.14
	n_wh	0.00	0.00	0.00	0.00	0.00	0.33
	PPL	0.08	0.01	0.11	0.01	0.00	0.00
	sent_bool	0.21	0.01	0.02	0.00	0.01	0.00
		Grammar	Spelling	Convention	Comprehensiveness	Vividness	Sentence structure

The six variables in the upper part are the features generated by three cross-modal matching methods (i.e., ALBEF, GSMN and SGRAF), and they are mainly used in the prediction of “Comprehensiveness”. Each method estimated a set of similarities with their models trained on different datasets or with multiple scales of parameters, and we chose the two most effective similarities in our experiments. The variables in

the lower part are the features generated in natural language processing. This graph shows the importance of the variables by which the machine predicts the score, and they all fit our conjectures. For example, the vividness of a sentence mainly depends on whether a lot of adjectives and adverbs are used, and when scoring the sentences structure, more attention is paid to the number of clauses and pronouns.

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